

Solution Manual: Cramér–Wold Device Exercises

Concise solution sketches emphasizing key ideas.

1. Proof

Show $X_n \Rightarrow X$ iff $a^T X_n \Rightarrow a^T X$ for every $a \in \mathbb{R}^k$. Use characteristic functions or uniqueness of probability measures.

2. Gaussian vectors

Every linear combination is Gaussian; identify its mean and variance limit.

3. Marginals vs joint convergence

Construct dependent pairs with identical marginals but varying dependence structure.

4. Multivariate CLT

Apply the univariate CLT to $a^T X_n$ and invoke Cramér–Wold.

5. Uniqueness

The limits of all projections determine the characteristic function of the limit law.

6. Vector of means

Linear combinations reduce to means of transformed variables.

7. Slutsky

Apply Slutsky to each projection $a^T X_n$.

8. Joint asymptotic normality

Verify asymptotic normality of every linear combination.

9. Tightness

Projection convergence alone is insufficient without a suitable tightness argument in general settings.

10. Continuous Mapping

Take $g(x) = a^T x$ and apply the theorem.

11. Vector Delta Method

Apply Cramér–Wold to $J_g(\theta)X_n$ and use the scalar Delta Method.

12. Multinomial proportions

Linear contrasts satisfy a CLT; conclude by Cramér–Wold.

Key Statement

A sequence X_n in $\mathbb{R}^k$ converges in distribution to X if and only if $a^T X_n$ converges in distribution to $a^T X$ for every vector a in $\mathbb{R}^k$.